

QTM 220 Midterm Solution: Fall 2023 (Sept. 21)

Problem 1 (4 points)

Suppose you have a sample $Y_1, Y_2, \dots, Y_n$ from a population of unknown distribution, with mean μ and standard deviation σ (and therefore variance σ^2). While you do not know the population distribution, you do know that each Y_i is independent and identically distributed (i.i.d.).

Consider the following estimator for μ : $\tilde{Y} = \frac{1}{n-2} \sum_{i=1}^n Y_i$.

a) What is $E[\tilde{Y}]$?

$$\begin{aligned} E[\tilde{Y}] &= E\left[\frac{1}{n-2} \sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n-2} E\left[\sum_{i=1}^n Y_i\right] \\ &= \frac{1}{n-2} \left(\sum_{i=1}^n E[Y_i]\right) \\ &= \frac{1}{n-2} \left(\sum_{i=1}^n \mu\right) \\ &= \frac{1}{n-2} n\mu \\ &= \frac{n}{n-2} \mu \end{aligned}$$

b) What is the bias of $E[\tilde{Y}]$ for μ ?

$$\begin{aligned}
E[\tilde{Y} - \mu] &= \frac{n}{n-2}\mu - \mu \\
&= \left(\frac{n}{n-2} - 1\right)\mu \\
&= \left(\frac{n}{n-2} - \frac{n-2}{n-2}\right)\mu \\
&= \frac{2}{n-2}\mu
\end{aligned}$$

c) Showing your work, calculate $Var[\tilde{Y}]$.

$$\begin{aligned}
V[\tilde{Y}] &= V\left[\frac{1}{n-2} \sum_{i=1}^n Y_i\right] \\
&= \frac{1}{(n-2)^2} V\left[\sum_{i=1}^n Y_i\right] \\
&= \frac{1}{(n-2)^2} \left(\sum_{i=1}^n V[Y_i]\right) \\
&= \frac{1}{(n-2)^2} \left(\sum_{i=1}^n \sigma^2\right) \\
&= \frac{1}{(n-2)^2} n\sigma^2 \\
&= \frac{n\sigma^2}{(n-2)^2}
\end{aligned}$$

d) Is $\tilde{Y}$ a *consistent* estimator for the population mean μ ? Why or why not? **Yes, both its bias and variance go to zero.**

Problem 2

Suppose we are studying a trait that is distributed differently in two different groups in the population. We call these groups the “low” group (characterized by Y_{low}) and the “high” group (characterized by Y_{high}). Y_{low} and Y_{high} have the following distributions:

$$Y_{low} \underset{i.i.d.}{\sim} N(\mu, \sigma^2)$$

$$Y_{high} \underset{i.i.d.}{\sim} N(2\mu, \sigma^2)$$

The parameter of interest is μ . Suppose we are able to conduct separate random samples with replacement from the groups of sizes n_{low} and n_{high} .

a) Consider this estimator for μ : $\hat{\mu} = \sum_{j=1}^{n_{high}} a_j Y_{j,high} - \sum_{i=1}^{n_{low}} b_i Y_{i,low}$

Find the expected value of this estimator, $E[\hat{\mu}]$.

$$\begin{aligned} E[\hat{\mu}] &= E\left[\sum_{j=1}^{n_{high}} a_j Y_{j,high} - \sum_{i=1}^{n_{low}} b_i Y_{i,low}\right] \\ &= E\left[\sum_{j=1}^{n_{high}} a_j Y_{j,high}\right] - E\left[\sum_{i=1}^{n_{low}} b_i Y_{i,low}\right] \\ &= \sum_{j=1}^{n_{high}} E[a_j Y_{j,high}] - \sum_{i=1}^{n_{low}} E[b_i Y_{i,low}] \\ &= \sum_{j=1}^{n_{high}} 2\mu a_j - \sum_{i=1}^{n_{low}} \mu b_i \\ &= \mu\left(2 \sum_{j=1}^{n_{high}} a_j - \sum_{i=1}^{n_{low}} b_i\right) \end{aligned}$$

b) Showing your work, calculate $Var[\hat{\mu}]$.

$$\begin{aligned}
V[\hat{\mu}] &= V\left[\sum_{j=1}^{n_{high}} a_j Y_{j,high} - \sum_{i=1}^{n_{low}} b_i Y_{i,low}\right] \\
&= V\left[\sum_{j=1}^{n_{high}} a_j Y_{j,high}\right] + V\left[\sum_{i=1}^{n_{low}} b_i Y_{i,low}\right] \\
&= \sum_{j=1}^{n_{high}} V\left[a_j Y_{j,high}\right] + \sum_{i=1}^{n_{low}} V\left[b_i Y_{i,low}\right] \\
&= \sigma^2 \sum_{j=1}^{n_{high}} a_j^2 + \sigma^2 \sum_{i=1}^{n_{low}} b_i^2 \\
&= \sigma^2 \left(\sum_{j=1}^{n_{high}} a_j^2 + \sum_{i=1}^{n_{low}} b_i^2 \right)
\end{aligned}$$